

DP014
Physics 1
Semester 1
Session 2024/2025
2 hours

DAQ SET B



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**PROGRAM DRAW A QUESTION
MODEL PHYSICS QUESTION PAPER
2 HOURS**

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

INSTRUCTIONS TO CANDIDATE:

The question paper consists of 7 question.

Answer **all** questions.

The use of non-programmable scientific calculator is permitted.

QUESTION	MARKS
1	
2	
3	
4	
5	
6	
7	
TOTAL	

*This question paper consists of **10** printed pages.*

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_o	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ϵ_o	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$
Atomic mass unit <i>Unit jisim atom</i>	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19} \text{ J}$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Constant of proportionality for
Coulomb's law
Pemalar hukum Coulomb

$$k = \frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

Atmospheric pressure
Tekanan atmosfera

$$1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$$

Density of water
Ketumpatan air

$$\rho_w = 1000 \text{ kg m}^{-3}$$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|--|---|
| 1. $v = u + at$ | 17. $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$ |
| 2. $s = ut + \frac{1}{2}at^2$ | 18. $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$ |
| 3. $v^2 = u^2 + 2as$ | 19. $s = r\theta$ |
| 4. $s = \frac{1}{2}(u + v)t$ | 20. $v = r\omega$ |
| 5. $v = u - gt$ | 21. $a_t = r \propto$ |
| 6. $v^2 = u^2 - 2gs$ | 22. $\omega = \omega_o + \alpha t$ |
| 7. $s = ut - \frac{1}{2}gt^2$ | 23. $\omega^2 = \omega_o^2 + 2\alpha\theta$ |
| 8. $p = mv$ | 24. $\theta = \omega_o t + \frac{1}{2}\alpha t^2$ |
| 9. $J = F\Delta t$ | 25. $\theta = \frac{1}{2}(\omega_o + \omega)t$ |
| 10. $J = \Delta p = mv - mu$ | 26. $\tau = rF \sin \theta$ |
| 11. $f_s \leq \mu_s N$ | 27. $\frac{Q}{t} = -kA \left(\frac{\Delta T}{L} \right)$ |
| 12. $f_k \leq \mu_k N$ | 28. $v_{rms} = \sqrt{\langle v^2 \rangle}$ |
| 13. $W = \vec{F} \cdot \vec{s} = Fs \cos \theta$ | 29. $v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$ |
| 14. $K = \frac{1}{2}mv^2$ | 30. $\Delta U = Q - W$ |
| 15. $U = mgh$ | |
| 16. $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$ | |

1. (a) A density of ethanol at room temperature is 0.795 g cm^{-3} . What is the value in kilograms per meter cubic?

[2 marks]

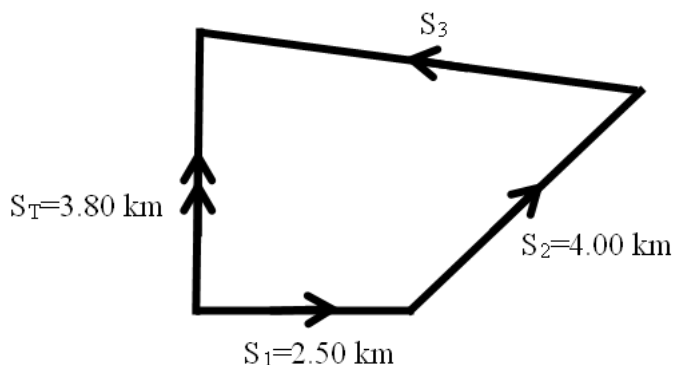


FIGURE 1

- (b) FIGURE 1 shows the route taken by a postal employee in a town. He drives a delivery truck 2.50 km to the east, S_1 , then 4.00 km north-east, S_2 and continue an additional distance of S_3 . If his total distance, S_T is 3.80 km directly north of his starting point, calculate the magnitude and direction of S_3 .

[6 marks]

2. (a)

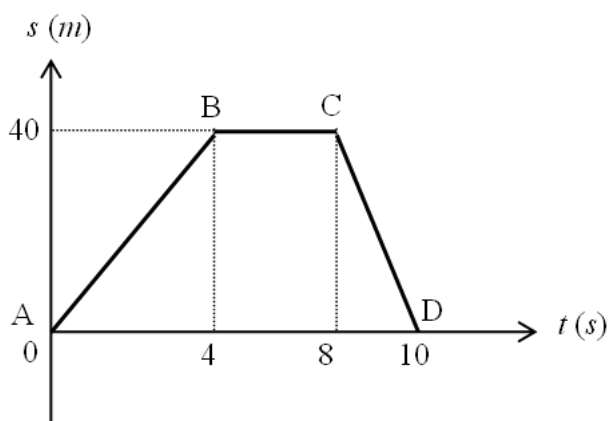


FIGURE 2

FIGURE 2 shows the motion of an object in a straight line. Calculate

- velocity of the object for the motions given by AB, BC and CD.
- total distance travelled until point D.
- average speed of the object.

[6 marks]

(b) A coin is flipped vertically upward with initial velocity is 8.0 m s^{-1} after leaving the thrower's hand. Calculate the

- maximum height, H of the coin.
- speed of the coin return to the thrower's hand.
- time flight that coin is in the air

[7 marks]

3. (a)

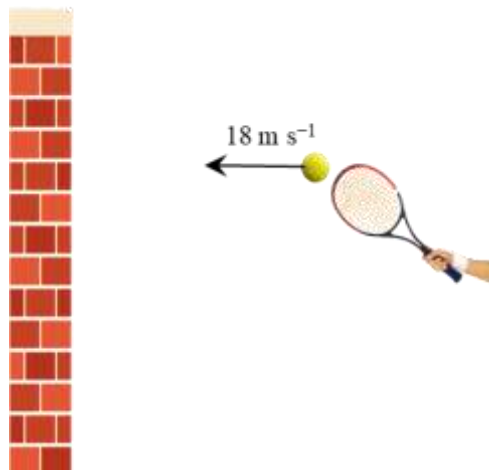


FIGURE 3

FIGURE 3 shows a tennis ball of mass 0.40 kg moving with a horizontal velocity of 18 m s^{-1} and hits a vertical wall. After 0.02 s , the tennis ball rebounds backward with the same velocity. Calculate the

- impulse exerted by the tennis ball on the wall.
- average force exerted by the tennis ball on the wall.

[4 marks]

(b)

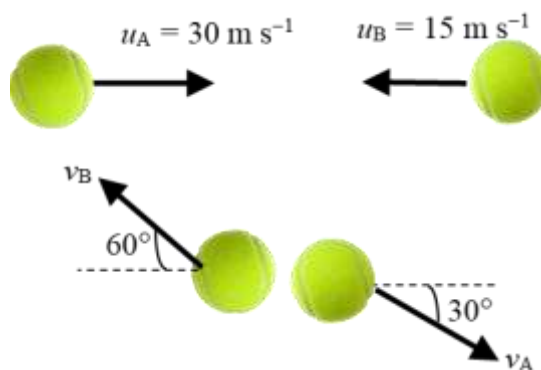


FIGURE 4

FIGURE 4 shows the two identical of tennis ball where ball A moves at 30 m s^{-1} and ball B moves at 15 m s^{-1} collide in the opposite direction. The two tennis balls are collided and bounced off each other. Calculate the final velocity of ball A and ball B after the collision.

[5 marks]

4. (a)

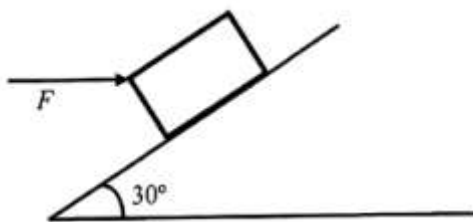
**FIGURE 5**

FIGURE 5 shows an object of mass 4 kg placed on a rough plane inclined at 30° with the horizontal. The coefficient of kinetic friction between the object and the plane surface is 0.25 . A constant horizontal force $F = 45 \text{ N}$ acts on the object and pushes it along the inclined plane with acceleration a .

- i. Sketch a free body diagram showing all the forces acting on the object.
- ii. Calculate the normal force of the object.
- iii. Calculate the acceleration of the object.

[8 marks]

(b) A load of 5000 kg is lifted vertically by a cable attached to a crane. The cable can support a maximum tension of $6.5 \times 10^3 \text{ N}$. Calculate the tension in the cable when

- i. the load is lifted vertically at constant speed of 3.0 m s^{-1} .
- ii. the load is lowered down at constant acceleration of 1.5 m s^{-2} .

[4 marks]

5. (a) A 25 kg girl is riding on a merry-go-round at a newly opened Funpark in Gamuda Luge Gardens. The merry-go-round has a 7 m diameter and rotates for 5 revolutions in 6 minutes. (i.e. the gravity can be neglected). Calculate the

- i. angular velocity of the girl.
- ii. centripetal force acts on the girl.

[4 marks]

(b)

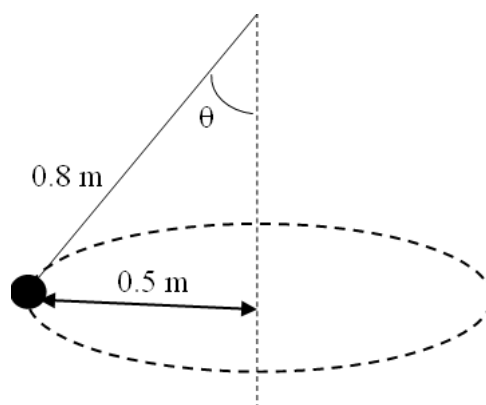
**FIGURE 6**

FIGURE 6 shows that an 85 g steel ball is attached to a 0.8 m length of string. It moves in a horizontal circle and the distance between the steel ball and the vertical line is 0.5 m. Calculate the

- i. angle, θ between the string and the vertical.
- ii. tension on string.
- iii. tangential speed of the steel ball.

[5 marks]

6. (a) A grinding wheel has a radius of 0.05 m is initially at rest. It then rotates until it reaches 250 rev min^{-1} within 8.0 s. Calculate

- i. the final tangential speed.
- ii. the angular acceleration of the wheel rotates during the time interval 8.0 s.
- iii. angular displacement the wheel rotates during the time interval 8.0 s.

[7 marks]

(b) i.

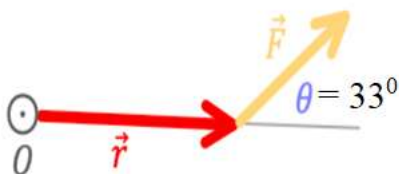
**FIGURE 7**

FIGURE 7 shows a 5.0 N force is applied at a distance, $r = 2.6 \text{ m}$ from the axis of rotation. Calculate the magnitude and direction of the torque, τ .

ii.

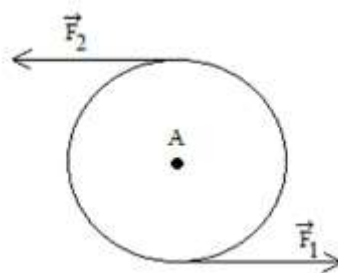
**FIGURE 8**

FIGURE 8 shows a solid object rotates around an axis at point A. Two forces, $F_1 = 30 \text{ N}$ and $F_2 = 45 \text{ N}$ is applied on the object which have a diameter of 4.0 cm . Calculate the resultant torque produced by the forces.

[5 marks]

7. (a) i. One face of an aluminium cube of edge 5 m is maintained at 60°C and the other end is maintained at 0°C . Calculate the amount of heat flowing through the cube in 2 s . The heat is flowing steadily along the cube.
(Thermal conductivity of aluminium = $209 \text{ W m}^{-1}^\circ\text{C}^{-1}$)

- ii. The rate of heat loss from a 10 cm thick slab of steel is 50 J s^{-1} . If the temperature difference of -10 K , calculate the area of the slab.
(Thermal conductivity of steel = $45 \text{ W m}^{-1} \text{ K}^{-1}$)

- iii. An insulated copper cube has a side of 16 cm . Two opposite surfaces of the cube have the temperature of 150°C and 100°C respectively. Sketch a label graph of temperature-distance of the cube **and** calculate the temperature gradient. (Thermal conductivity of copper = $399 \text{ W m}^{-1}^\circ\text{C}^{-1}$)

[8 marks]

- (b) i. Calculate the root mean square speed of carbon dioxide molecules at 350 K .
(Molar mass of carbon dioxide = 44 g mol^{-1})

- ii. The rms speed of nitrogen gas at a temperature of 20°C is 511 m s^{-1} . Calculate mass of an atom of the gas.

[4 marks]

- (c) i. A gas in a closed container is heated and causing the lid of the container to rise. The gas performs 6 J of work to raise the lid, so that the internal energy increases by 30 J. Calculate how much heat was added to the system.

ii.

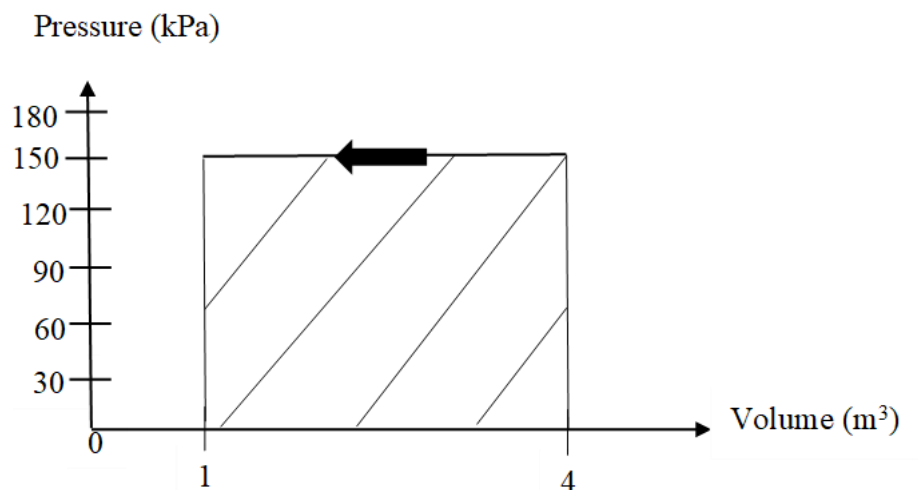


FIGURE 9

FIGURE 9 shows a gas in a container at a pressure of 150 kPa and a volume of 4 m³. Calculate the work done by the gas if it is compressed at constant pressure of 150 kPa.

[5 marks]

END OF QUESTION PAPER.